

Comprehensive list of PowerShell commands (cmdlets) commonly used in SQL Server Database Administration.

This includes **native SQL Server PowerShell (SQLPS / SqlServer module)** and **DBATools** — the most powerful and widely used PowerShell module for SQL DBAs.

1. Setup & Module Management

Purpose	Command
Install the SQL Server module	Install-Module -Name SqlServer
Import SQL Server module	Import-Module SqlServer
Install DBATools module	Install-Module -Name dbatools
Import DBATools module	Import-Module dbatools
List SQL modules installed	Get-Module -ListAvailable *sql*
Update DBATools	Update-Module dbatools

2. SQL Server Instance Management

Task	Command
Get all SQL Server instances on network	Get-DbalInstance -DiscoveryType Network
Connect to a SQL instance	Connect-DbalInstance -SqlInstance 'ServerName'
Get SQL instance info	Get-DbalInstanceProperty -SqlInstance 'ServerName'
Restart SQL Service	Restart-Dbaservice -ComputerName 'ServerName' -InstanceName 'MSSQLSERVER'
Get SQL services status	Get-Dbaservice -ComputerName 'ServerName'
Start/Stop SQL service	Start-Dbaservice / Stop-Dbaservice
Check SQL Server build version	Get-Dbabuildreference
Get SQL Server feature list	Get-Dbafeature -ComputerName 'ServerName'

3. Database Management

Task	Command
List all databases	Get-Dbadatabase -SqlInstance 'ServerName'
Create new database	New-Dbadatabase -SqlInstance 'ServerName' -Name 'TestDB'
Drop a database	Remove-Dbadatabase -SqlInstance 'ServerName' -Database 'TestDB'
Set database owner	Set-Dbadbowner -SqlInstance 'ServerName' -Database 'DBName' -TargetLogin 'sa'
Get database file info	Get-Dbadbfile -SqlInstance 'ServerName' -Database 'DBName'
Shrink database	Shrink-Dbadatabase -SqlInstance 'ServerName' -Database 'DBName'
Change DB state (Online/Offline/Readonly)	Set-Dbadbstate

4. Backup & Restore

Task	Command
Full backup	Backup-DbaDatabase -SqlInstance 'ServerName' -Database 'DBName' -Type Full -Path 'D:\Backup'
Differential backup	Backup-DbaDatabase -Type Differential
Log backup	Backup-DbaDatabase -Type Log
Restore database	Restore-DbaDatabase -SqlInstance 'ServerName' -Path 'D:\Backup\DBName.bak'
Verify backup	Test-DbaBackup -Path 'D:\Backup\DBName.bak'
Show backup history	Get-DbaBackupHistory -SqlInstance 'ServerName' -Database 'DBName'
Remove old backups	Remove-DbaBackup -Path 'D:\Backup' -OlderThan (Get-Date).AddDays(-7)

5. Login & Security Management

Task	Command
List all logins	Get-DbaLogin -SqlInstance 'ServerName'
Add a new login	New-DbaLogin -SqlInstance 'ServerName' -Login 'NewUser' -SecurePassword (Read-Host -AsSecureString)
Remove a login	Remove-DbaLogin -SqlInstance 'ServerName' -Login 'OldUser'
Export logins	Export-DbaLogin -SqlInstance 'ServerName' -Path 'C:\Logins.sql'
Import logins	Import-DbaLogin -SqlInstance 'ServerName' -Path 'C:\Logins.sql'
Sync logins between servers	Copy-DbaLogin -Source 'Server1' -Destination 'Server2'

6. Jobs, Schedules & Alerts

Task	Command
List SQL Agent jobs	Get-DbaAgentJob -SqlInstance 'ServerName'
Start a job	Start-DbaAgentJob -SqlInstance 'ServerName' -Job 'Backup Job'
Stop a job	Stop-DbaAgentJob -SqlInstance 'ServerName' -Job 'Backup Job'
Get job history	Get-DbaAgentJobHistory -SqlInstance 'ServerName'
Disable/Enable job	Set-DbaAgentJob -Enabled:\$false / -Enabled:\$true
Create new job	New-DbaAgentJob
Export jobs	Export-DbaAgentJob
Copy jobs between servers	Copy-DbaAgentJob -Source 'Server1' -Destination 'Server2'

7. Performance & Monitoring

Task	Command
Check CPU/memory usage	Get-DbaProcess -SqlInstance 'ServerName'
View active sessions	Get-DbaProcess -SqlInstance 'ServerName'
Kill a session	Stop-DbaProcess -SqlInstance 'ServerName' -Spid 52
Check wait stats	Get-DbaWaitStatistic -SqlInstance 'ServerName'
Get top queries	Get-DbaTopResourceUsage -SqlInstance 'ServerName'
Export performance data	Export-DbaDiagnosticData
Run DMV queries	Invoke-DbaQuery -SqlInstance 'ServerName' -Query "SELECT * FROM sys.dm_exec_requests"

8. Replication / AlwaysOn / HA

Task	Command
Get Availability Group info	Get-DbaAvailabilityGroup -SqlInstance 'ServerName'
Get replicas	Get-DbaAgReplica -SqlInstance 'ServerName'
Add database to AG	Add-DbaAgDatabase
Suspend/Resume AG DB	Suspend-DbaAgDbDataMovement / Resume-DbaAgDbDataMovement
Get AG listener info	Get-DbaAgListener
Test AG health	Test-DbaAvailabilityGroup
Get replication info	Get-DbaReplication

9. Maintenance & Utilities

Task	Command
Run maintenance script	Invoke-DbaQuery -File 'C:\Scripts\Maintenance.sql'
Check database consistency	Test-DbaDbIntegrity
Rebuild/Reorganize indexes	Invoke-DbaDbIndex
Update statistics	Update-DbaDbStatistic
Clean tempdb	Invoke-DbaDbShrink -Database tempdb
Apply patches	Install-DbaInstance
Generate documentation	Export-DbaInstance



10. Migration & Comparison

Task	Command
Copy databases between servers	Copy-DbaDatabase -Source 'Server1' -Destination 'Server2' -Database 'DBName'
Compare databases	Compare-DbaDbSchema
Compare logins	Compare-DbaLogin
Copy SQL Agent jobs	Copy-DbaAgentJob
Copy linked servers	Copy-DbaLinkedServer
Copy credentials	Copy-DbaCredential



11. Scripting & Query Execution

Task	Command
Run SQL query	Invoke-Sqlcmd -Query "SELECT name FROM sys.databases" -ServerInstance 'ServerName'
Run query from file	Invoke-Sqlcmd -InputFile 'C:\Scripts\Query.sql' -ServerInstance 'ServerName'
Export query results	Invoke-Sqlcmd ...
Execute multiple scripts	Invoke-DbaQuery -SqlInstance 'ServerName' -File 'C:\Scripts*.sql'



12. Auditing & Reporting

Task	Command
Get login audit data	Get-DbaLoginAudit
Generate server report	Export-DbaInstance -Path 'C:\Reports\'
Export database permissions	Export-DbaUserPermission
Audit backup history	Get-DbaBackupHistory
Audit job success/failure	Get-DbaAgentJobHistory



13. Useful General Commands

Command	Description
Get-Help <command>	Displays help for any command
Get-Command *dba*	Lists all DBATools commands
Update-Help	Updates local help files
Out-GridView	Displays results in GUI grid
Export-Csv	Saves results to CSV
ConvertTo-Json	Exports data in JSON format

DBATools PowerShell commands for SQL Server DBAs, which is absolutely the **#1 modern automation toolkit** for SQL Server administration. Let's go over everything you need — including **what DBATools is**, **how to set it up**, and **the most powerful DBA-focused commands** (with practical use cases).

⚡ 1. What Is DBATools?

DBATools is an open-source PowerShell module built by DBAs *for DBAs*, containing **700+ cmdlets** to manage, migrate, monitor, and automate SQL Server.

- ✅ Supports all SQL Server versions from **2000 to 2022**
- ✅ Works on **Windows, Linux, and Azure SQL**
- ✅ Automates patching, migration, backups, jobs, logins, AlwaysOn, etc.

🔧 2. Installation & Setup

Install DBATools from PowerShell Gallery (Run as Administrator)

```
Install-Module dbatools -Scope AllUsers
```

Import module

```
Import-Module dbatools
```

Verify installation

```
Get-Module dbatools
```

```
Get-Command -Module dbatools | Measure-Object
```

🔧 3. Core DBA Categories & Common Commands

Let's break down DBATools cmdlets by what a **SQL Server DBA** does daily 📌

💻 Instance & Service Management

Task	Command	Example
Get SQL instance info	Get-DbalInstanceProperty	Get-DbalInstanceProperty -SqlInstance Server01
Get SQL services	Get-Dbaservice	Get-Dbaservice -ComputerName Server01
Start/Stop/Restart SQL Service	Start-Dbaservice / Stop-Dbaservice / Restart-Dbaservice	Restart-Dbaservice -ComputerName Server01 - InstanceName MSSQLSERVER
Check SQL version/build	Get-Dbabuildreference	Get-Dbabuildreference -SqlInstance Server01
Apply patch/update	Install-DbalInstance	Install-DbalInstance -Path C:\PatchFolder

Database Management

Task	Command	Example
Get all databases	Get-DbaDatabase	Get-DbaDatabase -SqlInstance Server01
Create a database	New-DbaDatabase	New-DbaDatabase -SqlInstance Server01 -Name HRDB
Remove a database	Remove-DbaDatabase	Remove-DbaDatabase -SqlInstance Server01 -Database HRDB
Set database owner	Set-DbaDbOwner	Set-DbaDbOwner -SqlInstance Server01 -Database HRDB -TargetLogin sa
Change state (offline/online)	Set-DbaDbState	Set-DbaDbState -SqlInstance Server01 -Database HRDB -Offline
Shrink/Optimize DB	Invoke-DbaDbShrink	Invoke-DbaDbShrink -SqlInstance Server01 -Database TempDB

Backup & Restore Automation

Task	Command	Example
Backup full database	Backup-DbaDatabase	Backup-DbaDatabase -SqlInstance Server01 -Database HRDB -Type Full -Path D:\Backup
Differential backup	Backup-DbaDatabase -Type Differential	
Log backup	Backup-DbaDatabase -Type Log	
Restore database	Restore-DbaDatabase	Restore-DbaDatabase -SqlInstance Server02 -Path D:\Backup\HRDB.bak
Verify backup	Test-DbaBackup	Test-DbaBackup -Path D:\Backup\HRDB.bak
Get backup history	Get-DbaBackupHistory	Get-DbaBackupHistory -SqlInstance Server01 -Database HRDB

Security & Logins

Task	Command	Example
Get logins	Get-DbaLogin	Get-DbaLogin -SqlInstance Server01
Create new login	New-DbaLogin	New-DbaLogin -SqlInstance Server01 -Login MyUser -SecurePassword (Read-Host -AsSecureString)
Copy logins to another server	Copy-DbaLogin	Copy-DbaLogin -Source Server01 -Destination Server02
Export logins	Export-DbaLogin	Export-DbaLogin -SqlInstance Server01 -Path C:\Logins.sql
Import logins	Import-DbaLogin	Import-DbaLogin -SqlInstance Server02 -Path C:\Logins.sql

 SQL Agent Jobs

Task	Command	Example
Get SQL Agent jobs	Get-DbaAgentJob	Get-DbaAgentJob -SqlInstance Server01
Start/Stop job	Start-DbaAgentJob / Stop-DbaAgentJob	Start-DbaAgentJob -SqlInstance Server01 -Job 'Nightly Backup'
Get job history	Get-DbaAgentJobHistory	Get-DbaAgentJobHistory -SqlInstance Server01 -Job 'IndexMaintenance'
Copy jobs to another server	Copy-DbaAgentJob	Copy-DbaAgentJob -Source Server01 -Destination Server02
Export jobs	Export-DbaAgentJob	Export-DbaAgentJob -SqlInstance Server01 -Path C:\Jobs\

 Performance, Monitoring, and Health Checks

Task	Command	Example
Get active sessions	Get-DbaProcess	Get-DbaProcess -SqlInstance Server01
Kill session	Stop-DbaProcess	Stop-DbaProcess -SqlInstance Server01 -Spid 52
Get top queries	Get-DbaTopResourceUsage	Get-DbaTopResourceUsage -SqlInstance Server01
Check wait stats	Get-DbaWaitStatistic	Get-DbaWaitStatistic -SqlInstance Server01
Run full health check	Test-DbaMaxMemory, Test-DbaTempdbConfig, Test-DbaLastBackup	e.g. Test-DbaLastBackup -SqlInstance Server01

 Migration & Copying Between Servers

Task	Command	Example
Copy databases	Copy-DbaDatabase	Copy-DbaDatabase -Source Server01 -Destination Server02 - Database HRDB
Copy linked servers	Copy-DbaLinkedServer	Copy-DbaLinkedServer -Source Server01 -Destination Server02
Copy credentials	Copy-DbaCredential	Copy-DbaCredential -Source Server01 -Destination Server02
Copy logins/jobs/operators	Copy-DbaLogin, Copy-DbaAgentJob, Copy-DbaAgentOperator	
Copy all instance objects	Copy-DbaInstance	Copy-DbaInstance -Source Server01 -Destination Server02

AlwaysOn Availability Groups (AGs)

Task	Command	Example
Get AG info	Get-DbaAvailabilityGroup	Get-DbaAvailabilityGroup -SqlInstance Server01
Get replicas	Get-DbaAgReplica	Get-DbaAgReplica -SqlInstance Server01
Add DB to AG	Add-DbaAgDatabase	Add-DbaAgDatabase -SqlInstance Server01 - AvailabilityGroup AG1 - Database HRDB
Suspend/resume data movement	Suspend-DbaAgDbDataMovement / Resume-DbaAgDbDataMovement	
Test AG health	Test-DbaAvailabilityGroup	Test-DbaAvailabilityGroup -SqlInstance Server01

Reporting, Auditing & Documentation

Task	Command	Example
Export instance report	Export-DbaInstance	Export-DbaInstance -SqlInstance Server01 -Path C:\Reports
Get permission report	Export-DbaUserPermission	Export-DbaUserPermission -SqlInstance Server01 -Path C:\Permissions
Get backup summary	Get-DbaBackupHistory	
Generate HTML documentation	Export-DbaInstance -SqlInstance Server01 -Path C:\Reports -Format HTML	

Scripting & Query Automation

Task	Command	Example
Run SQL query	Invoke-DbaQuery	Invoke-DbaQuery -SqlInstance Server01 -Query "SELECT @@VERSION"
Run scripts from file	Invoke-DbaQuery -File 'C:\Scripts\Maintenance.sql'	
Run against multiple servers	Invoke-DbaQuery -SqlInstance Server01, Server02 -Query "SELECT name FROM sys.databases"	
Export result to CSV	Invoke-DbaQuery ...	Export-Csv 'C:\Output.csv'

4. Learn & Explore DBATools

Resource	Description
https://dbatools.io	Official documentation and tutorials
Get-Help *Db*	Lists all DBATools commands with help
Get-Command -Module dbatools	See all available cmdlets
Find-DbCommand -Tag Backup	Find commands by function (e.g., Backup, Job, Login)

5. Example: End-to-End DBA Automation Script

Connect to SQL Instance

\$instance = 'SQLPROD01'

Check database health

Test-DbadbIntegrity -SqlInstance \$instance

Backup all user databases

Backup-Dbadatabase -SqlInstance \$instance -Type Full -Path 'D:\Backups'

Check last backup time

Test-DbalastBackup -SqlInstance \$instance

Export logins

Export-Dbalogin -SqlInstance \$instance -Path 'D:\Scripts\Logins.sql'

Generate report

Export-Dbainstance -SqlInstance \$instance -Path 'D:\Reports\InstanceReport.html' -Format HTML

<https://www.sqldbachamps.com>

<https://github.com/PMSQLDBA/PraveenMadupu>

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